UCS303: Operating System

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Course Objective: To understand the role, responsibilities, and algorithms involved for achieving various functionalities of an Operating System.

Syllabus

Introduction and System Structures: Computer-System Organization, Computer-System Architecture, Operating-System Structure, Operating-System Operations, Process Management, Memory Management, Storage Management, Protection and Security, Computing Environments Operating-System Services, User and Operating-System Interface, System Calls, Types of System Calls, System Programs, Operating-System Design and Implementation, Operating-System Structure, Operating-System Design & Virtualization

Process Management: Process Concept, Process Scheduling, Operations on Processes, Inter process Communication, Multi-threaded programming: Multi-core Programming

Multithreading Models, Process Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Algorithm Evaluation.
Deadlock: System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.
Memory Management: Basic Hardware, Address Binding, Logical and Physical Address, Dynamic linking and loading, Shared Libraries, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table, Virtual Memory Management: Demand Paging Copy-on-Write, Page Replacement, Allocation of Frames, Thrashing, Allocating Kernel Memory.
File Systems: File Concept, Access Methods, Directory and Disk Structure, File-System Mounting, File Sharing, Protection, File-System Structure, File-System Implementation Directory Implementation, Allocation Methods, Free-Space Management.
Disk Management: Mass Storage Structure, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, Swap-Space Management, RAID Structure.
Protection and Security: Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of the Access Matrix, Access Control, Revocation of Access Rights, Capability-Based Systems, The Security Problem, Program Threats, System and Network Threats, User Authentication, Implementing Security Defenses, Firewalling to Protect Systems and Networks.
Concurrency: The Critical- Hardware, Mutex Locks, Semaphores, Classic Problems of Synchronization, Monitors.

Laboratory Work

To explore detailed architecture and shell commands in Linux / Unix environment, Understanding of OS virtualization simulate CPU scheduling, Paging, Disk-scheduling and process synchronization algorithms and simulate multithreading processing scenarios with a focus on energy-efficient scheduling Students has to submit a mini project.

Sample Mini Projects:

- Develop a basic task manager that monitors CPU and memory usage and suggests actions to save energy.
- Simulate efficient virtual memory management techniques for optimized performance in industrial systems.
- Modify a disk scheduling algorithm to reduce energy consumption.
- Create a user-friendly interface for visually impaired users to interact with an OS.
- Simulate an OS that distributes processes efficiently across multiple cores.

Course Learning Objectives (CLO)

After the completion of the course, the student will be able to:

1. Explain the basic of an operating system viz. system programs, system calls, user mode and kernel mode.
2. Select a particular CPU scheduling algorithms for specific situation, and analyze the environment leading to deadlock and its rectification.
3. Explicate memory management techniques viz. caching, paging, segmentation, virtual memory, and thrashing.
4. Understand the concepts related to file systems, disk-scheduling, and security, protection.
5. Comprehend the concepts related to concurrency.

Text Books

. Operating System Concepts, Silberschatz A., Galvin B. P. and Gagne G., John Wiley & Sons Inc., 9th ed, 2013.
. Operating Systems Internals and Design Principles, Stallings W., Prentice Hall 9th ed, 2018

Reference Books

. Understanding the Linux Kernel, Bovet P. D., Cesati M., O'Reilly Media, 3rd ed, 2006.
. Introduction to Operating System Design and Implementation: The OSP 2 Approach, Kifer M., Smolka A. S., Springer, 2007